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Development of an internet of things based fire detection and automatic extinguishing system for smart buildings

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***Abstract*— Fire detectors and extinguishing systems are used for many purposes such as discovering the onset of fire outbreak, reporting outbreaks to concerned parties, managing risk, and as well as notifying the fire department and controlling the negative outcome of fire. This paper focuses on early fire detection and an extinguishing system, which aims to measure and detect anomalies of some specific variables that we found insightful (such as temperature, gas leakage, smoke) which can cause fire outbreaks in homes, offices etc. Notifications are sent to the registered number or desired personnel in charge VIA a push bullet application software and WhatsApp messages with the help of an online API system to inform them about possible fire outbreak on their properties while they are away from the building and also triggers the extinguishing system to stop the anomalies before or during the outbreak. This fire alarm is developed with a Node MCU ESP8266 board which support a Wi-Fi module, and other passive and active components. The board is controlled by a software program. It is used to monitor and perceive the environment for the onset of fire outbreak through the flame, smoke or gas, as well as temperature sensors. It helps to detect the possibility of fire outbreak from the first few seconds while a warning is generated by the system through text notification alert to the numbers registered via the WI-FI module using internet of things; simultaneously a DC fan is activated to reduce and put out both the temperature and fire respectively. In the end deploying this system reduced the risk of actual fire occurrence by 90% which is a good starting point for future improvement on system efficiency of the developed technique.**

Keywords- Internet of Things (IOT) WI-FI module, appliction, software, board, fire detection, extinguishing, API system

I. INTRODUCTION

Fire outbreak is an extremely dangerous event hence there is the need to monitor the room condition of important infrastructure and send out notifications before anything disastrous occurs. Majority of houses in developing countries do not have fire alarm system fittings and this leaves fire outbreak being unattended which in turn leads to avoidable loss of properties and human lives in many situations [3,7]. The onset of fire outbreak can also occur when the residents or owners of properties are far away; therefore there is the need for developing a remote monitoring, alarm signalling, and control system for fire outbreaks.

In recent years, variety of IOT devices used in home automation for monitoring and control is now common. The detection and control of fire outbreaks is one of the major applications of home automation using IOT [12, 5]. The IOT based fire detection and control system monitors the whole infrastructure, detects the fire during the first few seconds, triggers a signalling alarm and notifies the end user as well as the fire control station about the fire. It also attempts to extinguish the fire.

The installation of fire alarm systems in buildings over the last few decades as increased a lot. It enhanced the safety of human life and properties through notifying building occupants using audio alarm [9; 1]. However, it is of importance to be able to monitor, predicts and control possible fire outbreak before it occurs and also be able to notify properties owner about any possible incidence of fire out-break on their properties while they are away from it and deploy possible measures to combat fire outbreak. This design sought to implement a programmable board which support IOT application via its on-board Wi-Fi module, which helps to send visual notifications via test to multiple registered numbers on it to inform them about the situation of things on their properties.

This model sets out to design a system which can detect flame, gas leakage, abnormal temperature and any other parameters that can predict or diagnose fire danger in an enclosed space; design a system that can take decisive actions depending on the status diagnosis and implement the design using an infrared flame sensor, gas sensor and temperature sensor that sends a signal to the Micro-Controller (NodeMCU) which triggers the relay to extinguish the fire automatically. The model also informs the property owners or authorities in charge using Push Bullet Application and WhatsApp application with the help of this IOT based Fire Detection & Controlling System Using NodeMCU ESP8266. Overll, this paper contributes significantly to the methods used as both fire alarm and extinguishing system.

II. LITERATURE REVIEW

The Internet of Things (IOT) is a system of interrelated computing devices, mechanical and digital machines, objects, animals or people that are provided with unique identifiers (UIDs) and the ability to transfer data over a network without requiring human-to-human or human-to-computer interaction [13]. A thing in the internet of things can be a person with a heart monitor implant, a farm animal with a biochip transponder, an automobile that has built-in sensors to alert the driver when tire pressure is low or any other natural or man-made object that can be assigned an Internet Protocol (IP) address and is able to transfer data over a network [11]. IOT technology is related to things which belong to the idea of automation, such as smart grid, smart appliances, and smart homes. These things enhance each other and they can be controlled through mediums such as the communication mediums. The main concept of a network of smart devices was discussed as early as 1982 [8]. A growing number of IOT devices are created for consumer use which include, Smart home (Home automation), Elder care; provide assistance for those with disabilities and elderly individuals, organizational applications; medical and healthcare, transportation, and V2X communications; in vehicular communication systems.

Also known as IOT acquires and analyze data from connected equipment, locations, and people. It's used in manufacturing, agriculture, maritime etc. Monitoring and controlling operations of sustainable urban and rural infrastructures likes bridged, railway tracks etc. is a key application of IOT. Others include; Metropolitan scale deployment, Energy management, environmental monitoring, living lab etc. The internet of military things (IOMT) is the application of technology in the military domain, take for instance the internet of battlefield things, and so on.

An IOT ecosystem comprises of web-enabled smart devices that use embedded systems, such as processors, sensors and communication hardware, to collect, send and act on data they acquire from their environments [10]. IOT devices share the sensor data they collect by connecting to an IOT gateway or other edge device where data is either sent to the cloud to be analyzed or analyzed locally. Sometimes, these devices communicate with other related devices and act on the information they get from one another. The devices do most of the work without human intervention, although people can interact with the devices, for example, to set them up, give them instructions or access the data [6;7].

Sensors are devices that detect and respond to some type of input from the physical environment. The specific input could be light, heat, motion, smoke, gas, temperature, or anyone of the great number of other environmental phenomena. The output is generally a signal that is converted to human-readable display at the sensor location or transmitted electronically over a network for reading or further processing. Sensors can be classified into;

a) Active sensors which produce output signal with the help of external excitation supply. The own physical properties of the sensor vary with respect to the applied external effect. They are therefore also called self-generating sensors [4].
b) Passive sensors which produce output signal without the help of external excitation supply. They do not need any extra any extra stimulus or voltage.
c) Analogue sensors which produces continuous signal with respect to time with analog output. Generally, analogue voltage is in the range of 0 to 5V or current is produced as the output. They measure physical parameters like temperature, pressure, displacement, etc.
d) Digital sensors produce discrete output signals. Discrete signals will be non-continuous with time and it can be represented in ‘bits’ for serial transmission and in ‘bytes’ for parallel transmission.

A fire alarm system basically consists of a monitoring device, control panel, signaling alarms devices, and suitable communicating medium. Majority of fire monitoring and control systems are developed to diagnose the outbreak of fire very early before total destruction of lives and property.

III. PROPOSED MODEL

The design of the IOT based fire detector and automatic extinguishing system was based on the use of various electrical components. The goal was to develop a fire monitoring and controlling device which senses gas, smoke, temperature and flame using various sensors that can senses the above listed parameters and take certain decision when the value read by the sensor is above the threshold values. The design specifications are discussed below.

A. Block Diagram and Schematic Diagram of the IOT Fire Alarm and Extinguishing System

Fig. 1 and 2 explains the features of the design. The first box powers the NodeMcu board and provides it enough power as well as to other components around it. The sensors boards read their respective function signals from the room and send the signals to the NodeMcuEsp8266. The Node Mcu development board used which was embedded with a wireless module programmed to receive signals from the digital or analogue pin of the sensor after which certain decisions were made based on the diagnosed condition. The buzzer and LED box are the type of responses expected from the development board to indicate physical signal for people to react to. The last box is the extinguisher that is being activated by the relay to suppress the fire outbreak.

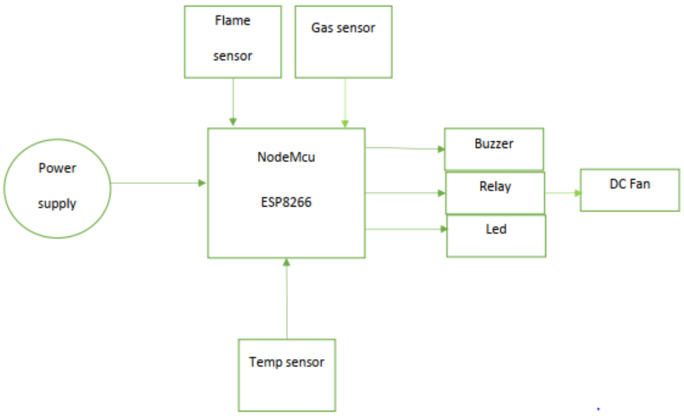

Fig.1. Block Diagram of the IOT Fire Alarm and Extinguishing System

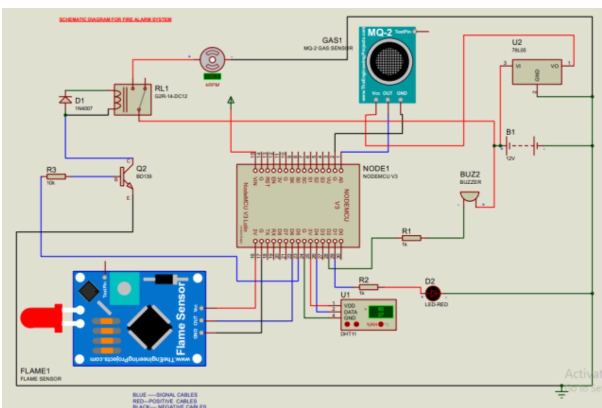

Fig. 2. Schematic Diagram of the Circuit

B. Flow Chart and Design Specifications

PCB Design

Fig. 3 shows the PCB design using the traditional method which involves the use of a copper board and ferric chloride for etching with dimensions as;

Width = 152.5mm

Height = 107.5mm

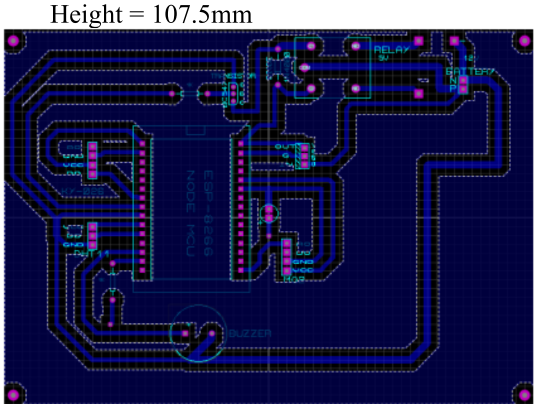

Fig. 3. PCB layout design

Flowchart

This flowchart explains how the code for the program runs and works simultaneously as shown in Fig. 4.

C. System Overview

The following sensors were used;

a. ESP8266; to send signals to the peripherals attached to it when it receives signal from the sensors through its analogue and digital pins. It is the central processing of the device.
b. DHT11; to measure the temperature and humidity in the room and this value are read through the digital pin of the sensor and the ESP8266 sends the value to the user in other to inform them the current status of the room.
c. MQ2 gas sensor; to diagnose the presentation and volume of harmful mixtures of gases in the air. This type of sensor is a metal oxide semiconductor which measures the volume and presentation of gases through a voltage divider circuit available at the sensor input terminal. When the sensing material comes in contact with any of the gases been investigated, the resistance of the material changes.

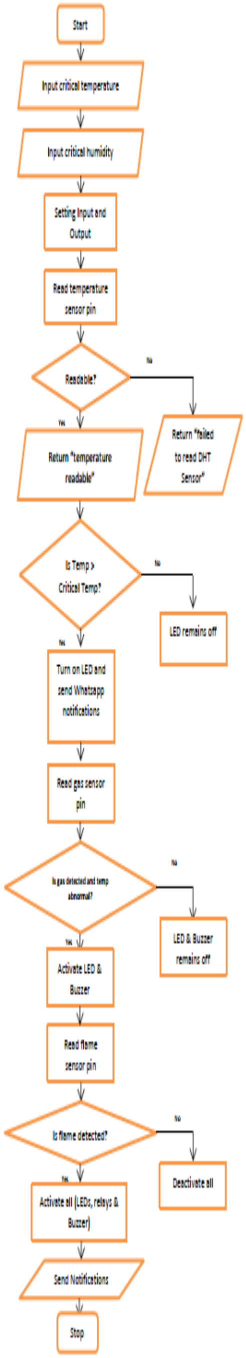

Fig. 4. Flow chart

d. KY-026 flame sensor; to diagnose the presence of infrared light emitted during fire outbreaks. This sensor is full of photo diodes and these diodes are mostly accurate in the detection of the spectral ranges of light that are created during the breakout fire leading to open flames. The sensor have variable resistor which makes it easy to adjust its responsiveness to both digital and analogue outputs. The main parts are the: infra-red receiver LED (usually about 5mm), indicator LEDs, resistors, male header pins dual differential comparator (LM393), and trimmer potentiometer (3296W).

After the sensors, the output transducers were also introduced. The first transducer employed is the buzzer. This is a device powered by a DC voltage, outputs sounds and conditions the output signals by converting electrical signals into sound signals. It is incorporated to be used as the system alarm. The buzzer used in this design is a 5V passive electromagnetic buzzer with 16Ω resistance value. It makes sounds when fire is detected to alert occupants in the building or people around the surrounding.

The second transduced used is the relay. The relay is a switch with small current ratings. The small current can latch on or off high currents. Relays have very useful applications small rated currents devices re required to activate high rated current devices. In order words these relays serve as both switches and amplifiers. The relay used in this project is a 5V relay and it is used as a switch to turn on the fan or leave the fan off. It is connected as normally open which closes and switch on the fan when it receives signals from the NodeMCU. A diode is connected in parallel with the relay in a reversed biased mode to prevent the flow of current and high voltage to be sent back into other components connected around it when the coil supply is interrupted.

The power supply of the NodeMcu was achieved using four lithium ion batteries were connected together in series to produce a 12V DC. The need for the supply of constant 5V was achieved using the voltage regulator 7805. This was used to provide a constant 5v to the NodeMcu board from the 12v DC source. A DC fan is powered by a 12V DC source; this fan serves as an extinguisher which helps to cool the room when temperature is rising beyond normal and also extinguish mild fire. In addition to the buzzer, an LED comes up when flame, gas or fire is being detected. It serves as a visual form of alarm.

IV. DESIGN IMPLEMENTATION, TESTING AND RESULT

A. *Prototype*

Fig. 5 shows the prototype of the designed IOT fire alarm system. This was done by integrating all individual blocks of figure one through the aid of the schematic circuit diagram and PCB as well as appropriate casing as shown in Fig. 6.

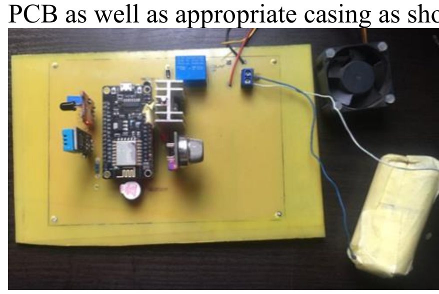

Fig. 5. Prototype of the designed IOT fire Alarm and Extinguishing System

B. 3-D Design

The 3-D design was design in this format in other to minimize cost and give it space to detect signals effectively as shown in Fig. 6.

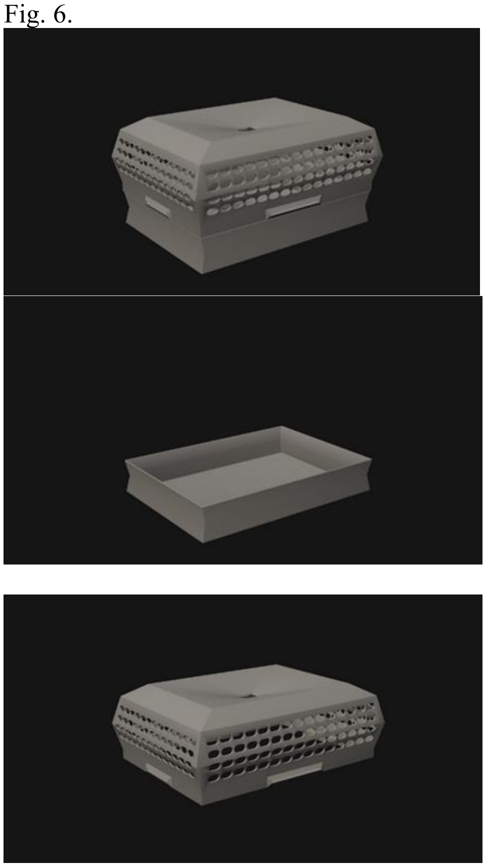

Fig. 6. 3-D Casing

Alert Notification Outputs

Fig. 7 shows the screenshots of notification messages received depending on the status of the condition monitored by DHT11, MQ2 and Flame Sensor.

Overall, each system components reacts to condition inputs as expected per time, 9 out of 10 times.

Fig.7a. Notification Messages

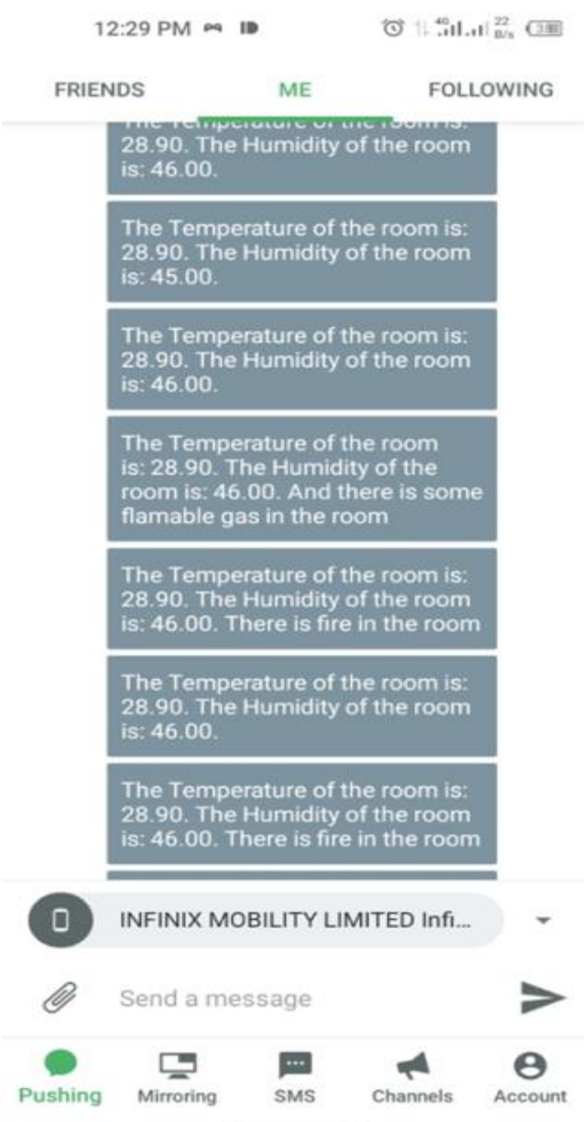

Fig. 7b. Notification Messages

V. CONCLUSION

This system was designed with the aim to develop a system which can serve as both a fire alarm and extinguishing system. The aim was achieved by the proposed system through the design of an Internet Of Things based fire alarm and extinguishing system that is capable of detecting the presence of fire and gases that can cause fire out-breaks and communicating to the concerned parties by sending notification to the users through Push-bullet Application and WhatsApp messages.

This work provided information about the design of electrical circuits, engineering, building and control on a relatively small scale. The project provided an opportunity to understanding fire alarm system, fundamentals and using the design and analysis tools. It also provided an opportunity for flexibilities in choosing electrical components and interactions.

REFERENCES

[1] Arduino Modules. https:// arduinomodules.info. Retrieved from https:// ardui nomodules. info: https:// arduinomodules.info/ky-026-flame-sensor-module/ 2021.

[2] Electrical Technokogy. (n.d.). Retrieved fromhttps://www.electricaltechnology.org//types-sensors-applications.html, 2018.

[3] Electrical4U. E4U. Retrieved from Electrical4U: https://www.electrical4u.com/diode-working-principle-and-types-of-diode/ 2021.

[4] EL-PRO-CUS. (n.d.). Retrieved from https://www.elprocus.com/, 2022.

[5] Evil Mad Scientist. (n.d.). Retrieved from Evil mad Scientist: https://shop.evilmadscientist.com/productsmenu/716

[6] A. S. Gills, "IoT Agenda". Retrieved from https:// internet of things agenda. techtarget.com/definition/Internet-of-Things-IoT?amp=1, 2021.

[7] PELONIs TECHNOLOGIES, INC., Retrieved from Pelonies Technologies, INC.:https://www.pelonistechnologies.com/blog/advantages-and-disadvantages-of-ac-fans-and-dc-fans, 2019.

[8] A. O. Akinwumi, "Improving System Reliability of Secondary Distribution Networks through Smart Monitoring" International Journal of Smart Security Technologies (IJSST) 9, vol.1, pp. 1-11, 2022.

[9] QUISURE (Electrical Supply Chain), Retrieved from https://www.quisure.com/blog/faq/what-is-the-working-principle-of-the-buzzer, 2020.

[10] QUISURE Electrical Supply Chain. Retrieved from https://www.quisure.com/blog/faq/what-is-the-working-principle-of-the-buzzer. 2022.

[11] ScienceDirect. (n.d.). Retrieved from Science Direct: https://www.sciencedirect.com/topics/chemistry/lithium-ion-battery, 2022.

[12] What is Industrial Automation | Types of Industrial Automation. (n.d.). Retrieved from https://www.electricaltechnology.org/2015/09/what-is-industrial-automation.html, 2022.

[13] O. Ojo and A. O. Akinwumi, "Design and implementation of a GSM based monitoring system for a Distribution Transformer" European Journal of Engineering and Technology Research, Issue 7 vol. 2, pp. 22-29, 2022.